Pollux marine drone

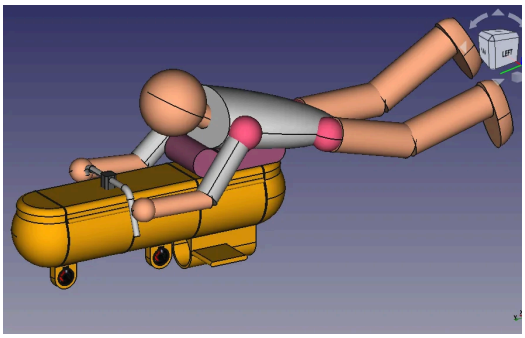
Project introduction

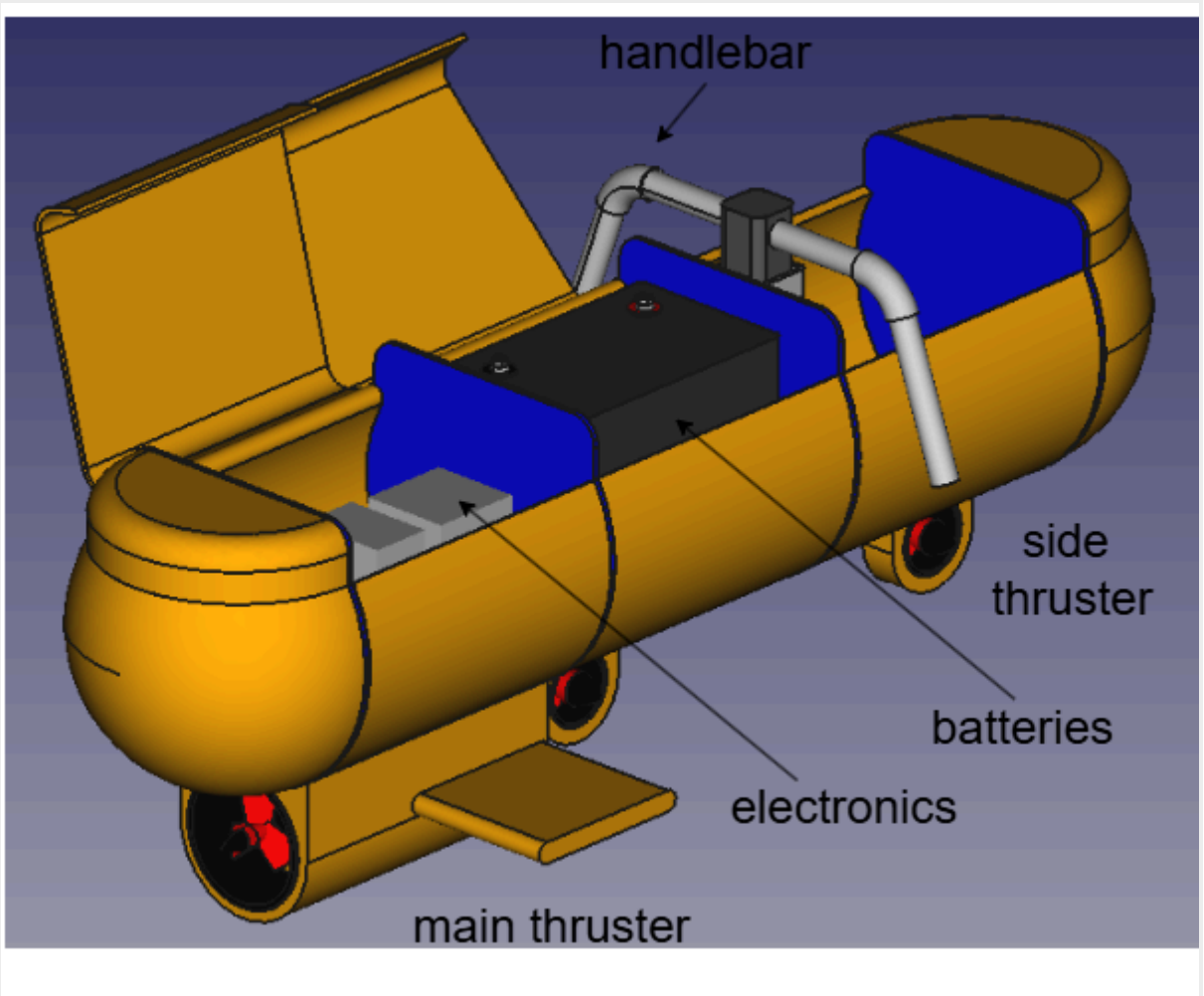
Last updated on March 3, 2026

Pollux is a surface marine drone that provides assistance to open sea swimmers as a kind of robotic “swimming buddy.” In autonomous mode, it follows the swimmer via a GPS beacon and responds to simple wrist-worn beacon commands (“Follow,” “Stop,” “Help”). It has enough thrust power to tow a tired or injured user to a safe location. It can also be driven like a sea scooter with manual controls for fun. The system is intended to work in four different modes: Follow, Waypoint, Rescue and Manual.

The Pollux architecture is open source and modular to encourage enthusiasts and students to add custom functionalities. The hull is cylinder shaped (approx 1m length, 20kg) and can house a battery bank and a thruster for up to 2 hours of continuous operation. It will be built with tools such as 3D printing, CNC machining, PCB design, Python physics analysis, Godot simulation, and OpenCV vision.

The architecture is composed of five main parts: the Hull, Communication, GPS, Thrust Control, and the Mind. The Vessel features a compact trimaran hull design for stability. The GPS continuously updates both the vessel’s and swimmer’s coordinates. The Communication module handles RF link commands from the swimmer’s wearable beacon. Thrust Control manages the physical actuators that drive the ship. Finally, the Mind integrates environmental data, user inputs, and system feedback to decide the appropriate course of action at every moment.


handlebar
side
thruster
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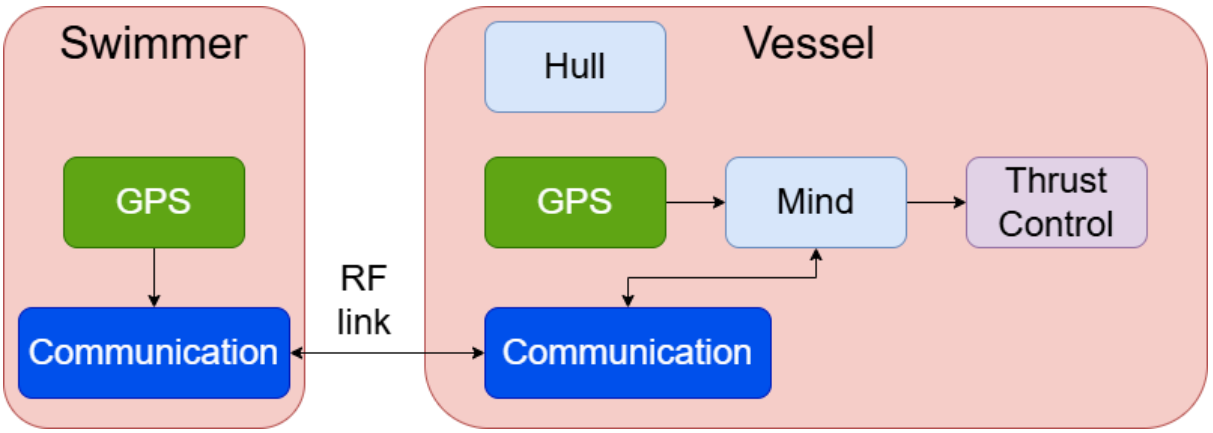
main thruster


Swimmer
GPS
Communication
RF
link
Vessel
Hull
GPS
Mind
Thrust
Control
Communication

Project function

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Wearable wristband

The wearable beacon contains one GPS chip and one RF link to transmit commands to Pollux. The design is intended to be light and comfortable, and the buttons are easy to reach - let's remember that training sessions in the open sea can be physically draining, so we want to avoid additional hassle to the user. The interface contains three buttons:

- **Mode**: Cycles between operational modes (Follow, Waypoint, Free).
- **Stop**: Stops for a rest
- **Rescue**: Initiates emergency return to a safe location.

The screen shows the Pollux's battery status, which is the most important parameter to watch during the mission.

Mode Descriptions

- **Follow mode:** Pollux trails behind the user within a 3m constant distance. Periodic alert sounds confirm its presence. The distance is long enough to avoid collision while close enough to assist the swimmer in case of an event.

- **Waypoint mode:** Pollux takes the lead and guides the user through different waypoints at a comfortable speed. The onboard Mind constantly monitors the swimmer position, adjusting the pace to keep the default distance (3m).

- **Free mode:** the vessel positions itself so that users can grab the handle bar and ride around for leisure. An LCD screen on the hull informs about the drone key parameters.

- **Rescue mode:** Pollux switches into autonomous mode, extracting the user to a predetermined safe location.

Project Parameters

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The most relevant parameters of the project are:

- **Battery status:** Determines the safe operating range - under normal circumstances, a 100% charge enables 1.5 to 2 hours of continuous run. When the battery falls under a critical threshold of 25%, the Mind will trigger an alert to the swimmer's wristband prompting a return to shore.
- **Thruster status**: The current and voltage supplied to the motor is monitored to ensure that they stay within safe operating limits (24V and 15A). Beyond these values, the motor risks overheating and having permanent damage.
- **Distance to the user.** A default value of 3m is good enough to keep during missions. Closer than that, it could lead to collision, while farther than that can delay the assistance in case of an event.
- **GPS coordinates:** Based on waypoints, swimmer and vessel coordinates, the Mind constantly adjusts the thruster power and the rudder angles to stay on course.

Hardware description

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The main hardware components are explained in this section.

The hull

It provides buoyancy and stability while housing the rest of the hardware (PCBs, actuators, sensors, etc.). It is 3D printed in PETG material and designed as modular containers that can be assembled like a Lego set. This modularity makes it easy for collaborators to replace containers and add new features, such as video cameras, extra thrusters or storage. Each container includes a servicing lid with a rubber gasket to against water infiltration.

Battery, Thrusters, and Rudders

These parts form the Thrust Control system that drives the vessel.

- **Battery:** Provides 24VDC up to 50A to onboard components. The charge is monitored by a BMS (Battery Management System). Voltage regulators generate secondary feeds at 3.3/5/12V. A fusebox protects against electrical faults.
- **Main Thruster:** A 24VDC 15A propeller powered by an ESC (Electronic Speed Control), providing backward and forward propulsion.
- **Rudders:** they steer the vessel left or right - controlled by servomotors.
- **Maneuvering thruster**: 12VDC 10A propellers that deliver lateral thrust for low-speed maneuvers.

Wristband

The wearable RF beacon integrates a GPS chip and a RF link for communication with the vessel. Its interface has three buttons for quick access (Mode, Stop, Rescue), an LCD screen to display critical information and a buzzer for audible alerts.

Onboard GPS

It is a GPS chip that continuously tracks the vessel position.

Onboard Comm

It is the vessel counterpart of the wristband RF link.

Onboard computer

An Arduino board that runs the main program, monitoring parameters and correcting the course according to mission needs. These subcircuits are part of it:

- **LCD screen:** Mounted on top of the hull, it displays general information and allows parameter personalization
- **Handlebar**. It contains the manual ride controls.
- **Navigation lights:** they increase visibility to nearby vessels and swimmers.
- **Buzzer**: It emits audible alerts to notify the swimmer and nearby vessels of important events.

Additional Sensors and Peripherals

Pollux supports expansion with sensors such as video cameras, LIDAR, thermal cameras and others. This approach encourages students and enthusiasts to experiment and learn while taking part in marine drone development.

Software code

GitHub - ramvqcraft/PolluxDrone: Pollux marine drone

Announcements

TBD

Assembling processes

TBD

Finished product display

TBD